

II BE EXAMINATION NOVEMBER, 2022

INFORMATION TECHNOLOGY

3ITRC3 Data Structures

Max Marks: 60

Duration: 3 Hrs

Note:

- IN EACH PROGRAM YOU MUST WRITE COMMENTS TO EXPLAIN THE CODE.
- Attempt any two subparts (a, b, c) of Question No. 1 to 3 and 5. From subparts of Question 4, attempt any four. Make and state assumptions wherever necessary. Write algorithm/code in readable form with proper indentation.

Q1.(a)	Define Data Structure. Explain the role of data structures in program design. Consider an array of n rows and n columns. Write a C/C++ function to rotate the array on column as shown in below example:	6																		
	<table border="1" style="display: inline-table; margin-right: 20px;"> <tr><td>C11</td><td>C12</td><td>C13</td></tr> <tr><td>C21</td><td>C22</td><td>C23</td></tr> <tr><td>C31</td><td>C32</td><td>C33</td></tr> </table> A 3x3 Array after rotation on column -> <table border="1" style="display: inline-table;"> <tr><td>C12</td><td>C13</td><td>C11</td></tr> <tr><td>C22</td><td>C23</td><td>C21</td></tr> <tr><td>C32</td><td>C33</td><td>C31</td></tr> </table>	C11	C12	C13	C21	C22	C23	C31	C32	C33	C12	C13	C11	C22	C23	C21	C32	C33	C31	
C11	C12	C13																		
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C31	C32	C33																		
C12	C13	C11																		
C22	C23	C21																		
C32	C33	C31																		
(b)	Compare Bubble Sort, Insertion Sort and Selection Sort on the basis of number of comparisons and number of swaps required in sorting N elements under best case and worst case.	6																		
(c)	Write a C/C++ program to implement single link list with following operations : (f) Insertion of element at first position (II) Insertion of element at last position (II) Insertion of element after p elements (IV) Removal of first element (V) Removal of last element (VI) Find smallest element from the list	6																		
Q2.(a)	Define stack. Write a program to implement stack using link list as container. How is stack used in handling recursive function calls?	6																		
(b)	Consider following definition for Ackermann function for non-negative integers : $A(m, n)$ defined as if $m=0$ then $A(m, n)=n+1$ if $m \neq 0$ but $n=0$ then $A(m, n) = A(m-1, 1)$ if $m \neq 0$ and $n \neq 0$ then $A(m, n) = A(m-1, A(m, n-1))$ Write a C/C++ function for computing above function. Draw recursion call tree or Box diagram for computing $A(4, 3)$ and compute how many times the function was called for computing the final result value.	6																		
(c)	Briefly Explain four applications of stack data structure in computing domain.	6																		
Q3.(a)	Write a program to implement a priority queue (using link list) of elements, where element is an integer number and the priority of element is number modulus 10 (i.e. priority = $n \% 10$, 0 highest and 9 lowest priority). Your program should have functions for enqueue and dequeue operation.	6																		